Final Year Project Report

Voice Guardian: Revolutionizing Mine Safety with AI- Voice Controlled Robot

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A report submitted in part fulfillment of the degree of

Bachelor of Engineering (Electrical Engineering)

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CERTIFICATE

Thises is written by **Aamir**, and **Aqdass Hayat** under supervision of **Dr. Abdul Baseer,** and endorsed by the entire thesis committee, has been submitted to and accepted by the of the head of Electrical Engineering Department, partially satisfying the requirements for the Bachelor of Electrical Engineering degree

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Project Supervisor HOD, Electrical Engineering

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Internal Examiner External Examiner

PREFACE

This thesis is being submitted in fulfilment of Sukkur IBA University's thesis requirement for the Bachelor of Electrical Engineering degree. The main focus of this research is the modification of Voice Guardian.

A team of two people completed this project, and the resulting thesis offers a thorough introduction to Voice controlled mobile robot. We would also want to express our appreciation to Dr. Abdul Baseer, our project supervisor, for all his help and encouragement.

Sukkur IBA University

Aamir (033-20-0027**)**

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DECLARATION

This report is based on our own work. Any other published or unpublished sources that have been used are properly acknowledged.

Word Count: 9612

- Aamir _____
- Aqdass Hayat _____

Date of Submission:

DEDICATION

With the grace of Allah Almighty and also the prayers of our We thank our parents and teachers for their support, encouragement, and unconditional love, teaching us to think clearly and motivating us to give our best in everything we do. We could not have achieved our goals without them and all who believe in the power of learning

ACKNOWLEDGEMENT

All praise be to Allah, the Almighty, the most Merciful and Compassionate, who granted us perseverance to undertake and accomplish this research work.

The work on this project has been an inspiring, often exciting, sometimes challenging, but always interesting experience. It has been made possible by many others who have supported us. So, we would like to take this opportunity to express our sincere gratitude to all those who have contributed to completing this project.

Firstly, we would like to thank our thesis adviser **Dr. Abdul Baseer**, Assistant Professor, Dept. of Electrical Engineering (EE), **Sukkur IBA University**, for his supportive guidance and feedback towards our project for its completion.

Finally, we wish to thank our parents for their undivided support and interest, who have inspired us and encouraged us to go our own way, without whom we would be unable to complete this project.

RESEARCH ETHIC FORM

I hereby affirm that the work presented in this thesis is original to me and was completed after my admission to the Bachelor of Engineering in Electrical Engineering program at Sukkur IBA University. No portion of this work has been included in any other thesis or dissertation submitted to this or any other institution for credit towards a degree, diploma, or other qualification. I agree to be responsible for ensuring that all procedures are carried out by the University's current research ethics guidelines. I have made every effort to foresee and mitigate all potential ethical, legal, and safety concerns that may come from this research, and I have gotten all necessary approvals before beginning.

Signature: _____

Date: _____

ABSTRACT

The project aims to allow user to control mobile robot using voice commands based on English and Urdu language datasets. The goal is to make robot control more accessible using voice, focusing on mine safety.

Our project uses machine learning techniques to accurately recognize and understand voice commands. A microphone on the robot captures the user's voice, which is then processed in real-time to direct the robot's movements and actions. The received samples of audio are converted to a spectrogram by using a short-time Fourier transform then the spectrogram enable CNN to perform feature extraction and pattern recognition. CNN model is trained on a dataset of voice commands it process spectrograms and performs prediction with good accuracy. It is then transferred learned to the Urdu data set, with the limited data set. Our system works in real time on Raspberry Pi 4 without relying on external servers. It has low latency and high reliability. Techniques like data augmentation and model pruning were used to enhance performance within hardware constraints.

The project is aimed to work on fully functional voice-controlled system that allows users to navigate the mobile robot and perform tasks using voice commands. It terms of applications it is focused on mine safety, providing a hands-free control mechanism to navigate hazardous environments, monitor conditions, and assist in emergency response. Additionally, the system can be applied in personal assistance, healthcare services and industrial automation. The successful implementation demonstrates the potential for deploying advanced machine learning algorithms in small, affordable hardware, making technology more accessible and convenient for critical safety applications and everyday tasks.

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CHAPTER 1

INTRODUCTION

The "Voice Guardian" project aims to revolutionize the operation of mobile robots. Traditional methods of controlling robots usually include complex programming or manual interfaces, which can be difficult for users lacking technical expertise. This project seeks to overcome these challenges by creating an innovative voice-controlled control system based on English and Urdu languages. This new system is is aimed to enhance, the efficiency, accessibility, and safety of mobile robots, making them more user-friendly and easier to operate. By taking advantage of natural language processing and advanced voice recognition algorithms, our system interprets user commands and translates them into executable actions for the robot. This approach allows users to control robots using simple voice commands, significantly reducing the complexity of robot operation. Moreover, the project is designed to be adaptable and scalable, by enabling integration with various mobile robot platforms across different industries. These industries include healthcare, agriculture, the power sector, the Internet of Things, robotics, and artificial intelligence.

1.1 Background

The foundation of this project lies in the intersection of voice recognition technology and mobile robotics. Traditional methods of controlling mobile robots typically involve using remote controls, joysticks, or computer interfaces. These methods require users to have a solid understanding of the robot's programming and operational protocols. For instance, operating a robot to navigate a specific path or perform a particular task often necessitates detailed programming instructions and precise manual inputs. This level of complexity can be a significant barrier, particularly for users in industries where technical skills may not be prevalent. Manual interfaces, while effective in precise control, but have certain limitations. They can be cumbersome and time-consuming, requiring constant attention and adjustments from the user. This can lead to inefficiencies, especially in dynamic environments where quick and adaptive responses are necessary. Moreover, manual control can be physically demanding, especially over prolonged periods, potentially leading to user fatigue and errors.

Voice recognition technology has seen significant advancements due to progress in artificial intelligence and NLP. Studies have showm an effectiveness of deep learning(DL) models, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), as these are achieving high accuracy in voice recognition tasks. These advancements enable the development of robust systems capable of understanding and executing natural language commands. Existing research has explored various aspects of Voice Guardians. For instance, a study by Garcia et al.

(2020) highlighted the effectiveness of deep learning models in voice recognition tasks. Another study by Gupta et al. (2019) examined transformer-based models for natural language understanding and speech-to-text conversion. Additionally, Nguyen et al. (2019) presented a voice-controlled service robot capable of performing tasks such as object retrieval and room navigation.

1.2 Motivation

The main motivation for this project is to enhance the accessibility and usability of mobile robots, secondly setting stage for Urdu language-based control. By leveraging natural and intuitive voice interfaces, the project aims to make robot control more user-friendly, particularly in critical applications like mine safety. The system's ability to navigate hazardous environments, monitor conditions, and assist in emergency responses can significantly improve safety and efficiency in these settings. Additionally, the project has potential to revolutionize different other industries, including healthcare, agriculture, and industrial automation. By providing an intuitive control mechanism, the project aims to increase the adoption of mobile robots in everyday tasks and critical operations.

1.3 Problem-Solving

The project addresses several key issues in mobile robot control it addresses complexity in control methods as robot control methods require significant technical expertise along with this it addresses the factor of limiting their usability, as the voice-controlled system simplifies control, and by making it in access to a broader range of users and along with this it also set stage for Urdu language voice control the area where there is very less amount of work.

In environments such as mines, where manual control can be hazardous, the voice-controlled system provides a hands-free mechanism for controlling robots, enhancing safety and the system ensures real-time processing and execution of commands, which is crucial for applications in dynamic and potentially dangerous environments along with this robust algorithm for noise cancellation are implemented to ensure the accuracy of voice recognition, even in noisy environments.

1.4 Aims and Objectives of Project

1.4.1 Aim:

This project aims to develop an advanced Voice Guardian that can perform different tasks based on natural language voice commands and to introduce work on Urdu language voice recognition.

1.4.2 Objectives

- Design a Mobile Robot Platform: The primary objective is to design a mobile robot platform that is suitable for implementing voice control and capable of navigating in different environments.
- Voice Recognition Implementation: Develop a robust voice recognition system capable of accurately interpreting natural language voice commands.
- Integration of Voice Control: Integrate the voice recognition system with the mobile robot platform to enable seamless and real-time control through voice commands.
- Task Execution: Implement functionalities to allow the robot to execute various tasks based on voice commands, such as navigation, object detection, and interaction with the environment.
- User-Friendly Interface: Ensure that the voice control system offers a user-friendly interface, making it easy for users to interact with the robot and issue commands naturally.
- Performance Optimization: Optimize the performance of the voice control system to minimize latency and improve the overall responsiveness of the robot.
- Testing and Validation: Conduct extensive testing and validation to ensure the accuracy and reliability of the voice control system under various conditions and scenarios.
- Performing transfer learning in the Urdu language.
- Adaptability and Scalability: Ensure the system is designed to be adaptable and scalable, enabling for future integration with different robot platforms and extending functionality for diverse applications.
- Safety Measures: Implement safety measures to prevent unintended actions and ensure the robot operates safely in different environments.
- Documentation and Reporting: Thoroughly document the design, development, and implementation process, along with performance evaluation results, for a comprehensive project report.

1.5 Applications

1.5.1 Mine Safety

The primary application of this system is in mine safety. Voice Guardians can be upgraded to navigate hazardous environments, monitor conditions, and assist in emergency responses, providing a hands-free control mechanism that enhances safety [16] [17]. These robots can be

remotely controlled using voice commands, allowing operators to maintain a safe distance from dangerous areas while still effectively managing operations.

1.5.2 Personal Assistance

Voice-controlled robots can be upgraded to serve as personal assistants, helping individuals with tasks such as scheduling, reminders, and information retrieval. This application is particularly beneficial for elderly or disabled individuals who may have difficulty using manual controls. By responding to voice commands, these robots can provide a convenient and accessible way for users to manage their daily activities and enhance their independence [19].

1.5.3 Industrial Automation

In industrial settings, voice-controlled robots can significantly enhance efficiency by performing tasks such as equipment monitoring, maintenance, and logistics management. These robots can reduce the need for manual intervention, thereby increasing productivity and minimizing the risk of human error. Operators can use voice commands to direct robots to perform specific tasks, streamline workflows, and monitor critical systems.

1.5.4 Healthcare

Voice-controlled robots have significant potential in healthcare. They can assist medical professionals with tasks such as patient monitoring, medication delivery, and telemedicine services [20]. By improving operational efficiency and reducing the workload on healthcare staff, these robots can enhance patient care. For example, a doctor can use voice commands to direct a robot to check vital signs, deliver medications, or facilitate remote consultations, allowing for more efficient patient management.

1.5.5 Agriculture

In agriculture, voice-controlled robots can be deployed for precision farming and crop monitoring. These robots can navigate fields, monitor crop health, and assist in resource management, leading to increased efficiency and reduced resource usage [18]. Farmers can issue voice commands to control the robots, enabling them to perform tasks such as planting, irrigation, and harvesting with greater accuracy and less physical effort.

1.6 Literature Review

The literature review provides an in-depth analysis of existing research, publications, and advancements related to Voice Guardians. This review aims to understand the current state-of-the-art in voice recognition technology, mobile robot design, and the integration of voice control in robotic systems. By examining the literature, we can identify key trends, challenges, and potential areas of improvement for our project on "Voice controlled Mobile Robot”

- Voice Recognition Technology: Voice recognition has witnessed significant advancements in recent years, driven by the rapid progress in artificial intelligence and natural language processing. Research by Garcia et al. (2020) demonstrated the effectiveness of deep learning models, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), in achieving high accuracy in voice recognition tasks. Additionally, Gupta et al. (2019) explored the use of transformer-based models for natural language understanding and speech-to-text conversion, showing promising results for real-time applications.
- Mobile Robot Design: The design of mobile robots has evolved to accommodate a diverse range of applications and environments. Literature by Smith et al. (2018) discussed the development of agile and compact robotic platforms that can navigate in confined spaces, making them suitable for indoor environments like warehouses and hospitals. Moreover, research by Wang et al. (2021) explored the integration of sensor fusion techniques, combining visual and depth sensors, to enhance robot perception and facilitate smooth navigation.
- Voice Guardians: Several studies have focused on the integration of voice control in mobile robots to enable intuitive and hands-free operation. Nguyen et al. (2019) presented a voice-controlled service robot capable of performing tasks like object retrieval and room navigation based on voice commands. Similarly, Zhang et al. (2020) designed a voice-controlled autonomous vehicle for outdoor environments, highlighting the potential of voice commands in complex tasks like route planning and obstacle avoidance.
- Challenges and Considerations: Despite the progress in Voice Guardians, several challenges persist. Literature by Li et al. (2017) highlighted issues related to ambient noise interference, which can impact the accuracy of voice recognition systems. Additionally, ensuring the safety of voice-controlled robots in dynamic environments remains a critical concern, as discussed by Jones et al. (2019). Addressing these challenges requires robust algorithms for noise cancellation, context-awareness, and human-robot interaction.

The literature review demonstrates the potential of Voice Guardians in various domains, including healthcare, agriculture, logistics, and more. It emphasizes the need for advanced voice recognition technology, efficient mobile robot design, and robust human-robot interaction to create a successful voice-controlled control system. By building upon the insights from existing research, our project aims to contribute to the development of a cutting-edge Voice Guardian capable of revolutionizing industries through intelligent and intuitive control mechanism

CHAPTER 2

METHODOLOGY

2.1 Voice Recognition Overview

Voice recognition enables a system to recognize and interpret human voice or speech and convert it into text or commands. This technology leverages various algorithms and models to process and understand spoken language, facilitating lots of applications from virtual assistants to automated customer service systems [1].

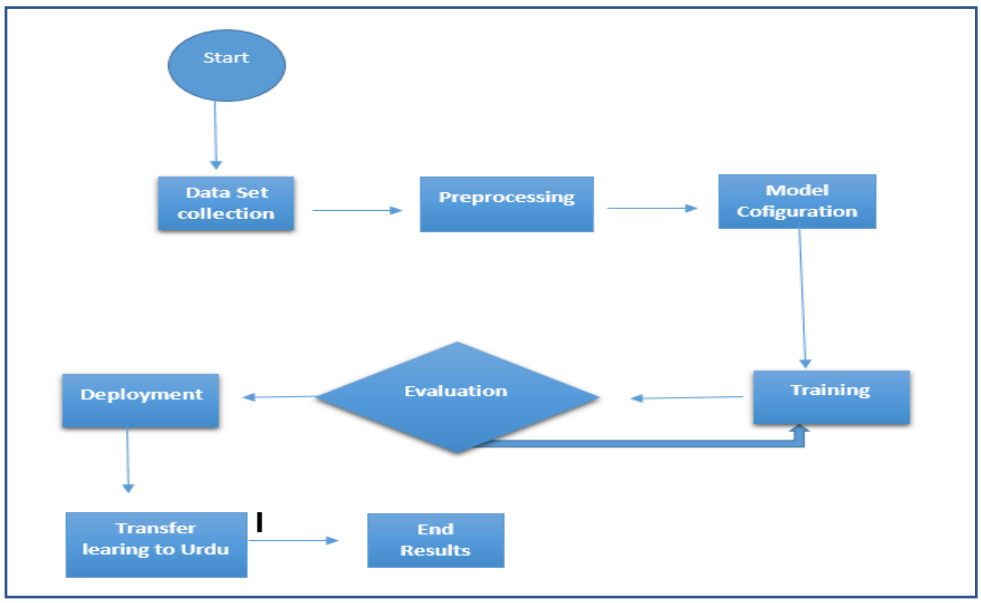


Fig. 1.1. Project Methodology.

2.1.1 Audio Capture

The process starts with capturing audio input using a microphone. The quality of the microphone and the environment (noise levels, acoustics) can significantly impact the accuracy of voice recognition [3].

2.1.2 Preprocessing

The captured audio signals are preprocessed to enhance the quality and reduce noise. This step involves techniques such as normalization, noise reduction, and filtering to improve the clarity of the speech signals [4].

2.1.3 Feature Extraction

In this step, the audio signals are converted into a set of features that can be used for further analysis. Common techniques include the STFT, Mel-Frequency Cepstral Coefficients and spectrograms. These features capture the essential characteristics of the speech signal [5].

2.1.4 Acoustic Modeling

Acoustic models represent the relationship between audio signals and phonetic units of speech. Hidden Markov Models (HMMs) and Deep Neural Networks (DNNs) are commonly used to model these relationships. These models help in mapping the audio features to the most likely phonemes or words [1].

2.1.5 Language Modeling

Language models predict the likelihood of a sequence of words. They help in improving the accuracy of speech recognition by considering the context and probability of word sequences. N-grams and Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM) networks, are often used for this purpose [6].

2.1.5 post-processing

The final output is further processed to improve readability and accuracy. This may include correcting common errors, applying grammatical rules, and formatting the recognized text [1].

2.2 Deep learning

Deep learning has made voice recognition much better. It uses special models like CNNs, RNNs (like LSTMs), and Transformers. These models turn sounds into features and then into words. The process starts with changing audio into spectrograms, which show frequencies over time. Features are then extracted from these spectrograms. Next, the models are trained on large datasets of speech and text. During use, these trained models turn new audio into text. CNNs help find features in the audio, RNNs understand the sequence of sounds, and Transformers help predict the right words. This makes voice recognition more accurate and scalable. But it needs a lot of data and computer power. Also, making sure it works well with different accents and noises is still a challenge. Overall, deep learning has greatly improved how we recognize and process speech [6].

2.3 Spectrogram

Spectrograms are visual representations that depict how the spectral characteristics of a signal change over time. They are essential tools in fields like speech processing, music analysis, and telecommunications, providing insights into the frequency content of signals in both the time and frequency domains [7]. The creation of spectrograms typically involves using the Short-Time Fourier Transform (STFT), which applies the Fourier Transform to small, overlapping segments of the signal. This process breaks down the signal into its constituent frequencies at different points in time, allowing for a detailed examination of how these frequencies evolve throughout the signal [8].

In practical terms, the STFT divides the signal into short windows, each of which undergoes a Fourier Transform to extract the frequency components present within that window. The size of the window used in STFT affects the trade-off between time and frequency resolution: smaller windows provide better time resolution by capturing rapid changes in the signal over time, but at the expense of poorer frequency resolution, making it harder to distinguish closely spaced frequencies. Overlapping these windows (typically by 50%) helps create a smooth transition

between adjacent segments, ensuring a continuous view of how frequencies change over time without abrupt shifts in the spectrogram display [8].

Each point in the spectrogram represents the intensity or amplitude of a particular frequency bin at a specific time, often depicted using a color or brightness scale. Stronger frequencies exhibit higher amplitudes, appearing brighter or more intense in the spectrogram. This visual representation aids in analyzing and interpreting complex frequency patterns within the signal, which would be challenging to discern from the raw waveform alone. In applications such as speech recognition, spectrograms serve as crucial inputs to deep learning models, enabling them to process and understand spoken language by capturing nuances like pitch variations, formants, and other spectral features that characterize different sounds and phonemes [10].

Overall, spectrograms are powerful tools for studying the temporal and spectral characteristics of signals, offering detailed insights into how signals vary over time across different frequencies. Their ability to visually represent complex frequency dynamics makes them indispensable in various scientific and engineering disciplines where understanding signal behavior and pattern recognition are paramount.

2.3.1 Advantages

Spectrograms allow for robust representation of speech signals, capturing nuances such as pitch, intensity, and formants. They enable the development of models that can recognize speech accurately across different speakers, accents, and environments [9].

Spectrograms excel in providing a comprehensive visual representation of a signal's frequency content over time, offering insights into complex frequency patterns and signal dynamics. This makes them valuable for tasks requiring detailed analysis of sound evolution and identification. In contrast, MFCC (Mel-Frequency Cepstral Coefficients) and LPC (Linear Predictive Coding) focus on extracting specific features suited for speech recognition: MFCCs for noise robustness and discriminative power, and LPC for modeling vocal tract characteristics and formats. Spectrograms' ability to capture a broad frequency range and temporal context makes them advantageous in applications like acoustic event detection and music analysis, where understanding both time and frequency components is critical as shown in Fig [1.1]. Ultimately, the choice between

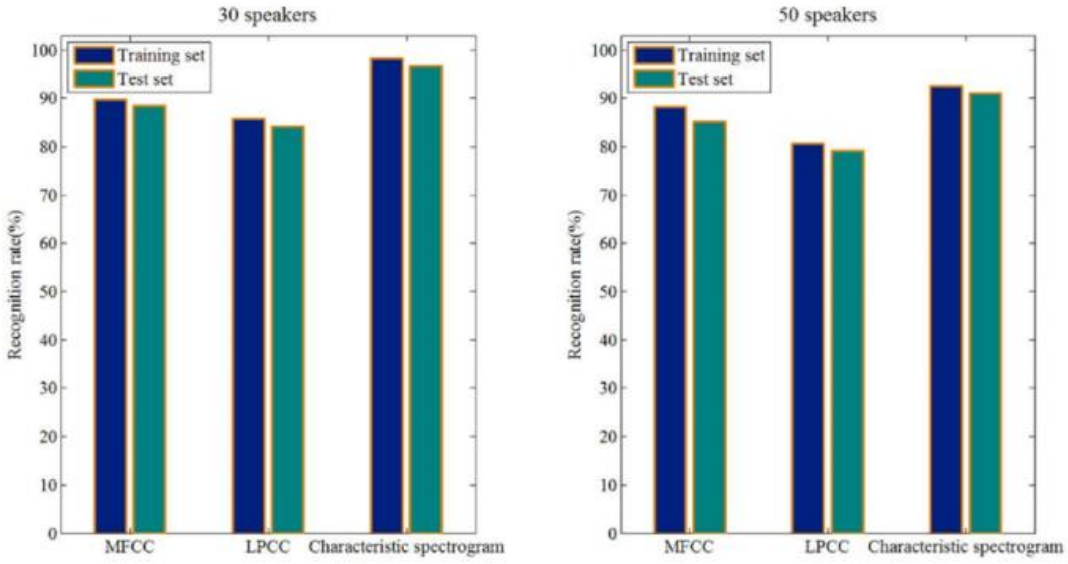


Fig. 2.2. Performance comparison for three different feature extraction methods [6]

spectrograms, MFCC, and LPC depends on the specific needs of the application, such as noise robustness, computational efficiency, and the level of detail required for analysis [10].

2.4 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a type of deep learning model specifically designed for tasks involving grid-like data such as images, spectrograms, and other structured inputs. They are widely used in fields like computer vision, speech recognition, and natural language processing due to their ability to automatically learn hierarchical representations of data [11].

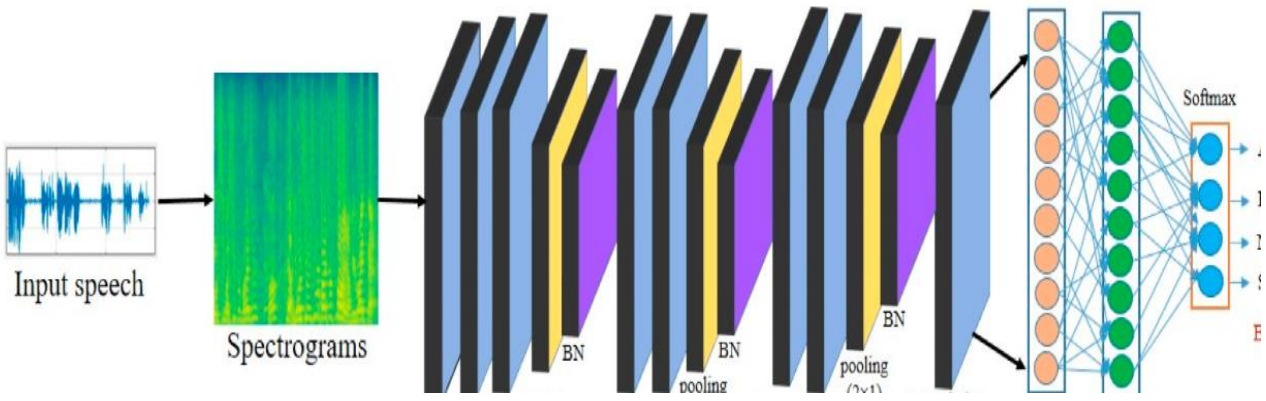


Fig. 2.3. Deep Frequency Features using CNN [5]

2.4.1 Structure and L

These layers apply fi ; the network to identify features sucl extract specific characteristics from tl ied Linear Unit),

Fig. 2.4. Block Diagram of Temperature SensorFig. 2.5. Deep Frequency Features using CNN [5]

introduce non-linearity into the network, enabling it to learn complex data relationships. Pooling layers (e.g., max pooling, average pooling) reduce the spatial dimensions of the data by summarizing the presence of features within sub-regions. This enhances the robustness and spatial invariance of the learned features. Towards the end of the network, fully connected layers integrate the features extracted by the convolutional and pooling layers to make final predictions [12]

2.4.2 Convolutional Operation

The convolutional operation involves sliding a small filter (kernel) over the input data and computing dot products to generate feature maps. Each filter captures different aspects of the input, such as edges or textures. This operation is illustrated by visualizing how filters slide over an image or spectrogram, showing the resulting feature maps that highlight specific patterns learned by the network [13].

2.4.3 Hierarchical Feature Learning

CNNs automatically learn hierarchical representations of data. Lower layers typically capture low-level features like edges and textures, while deeper layers combine these features into more abstract and complex patterns relevant to the task [11].

2.4.4 Training and Optimization

CNNs are trained using backpropagation with gradient descent, where weights are adjusted iteratively to minimize the difference between predicted and actual outputs (loss function).

2.4.5 Applications

CNNs are applied in diverse applications such as image classification, object detection, facial recognition, and speech recognition. For speech recognition, include spectrogram examples with overlaid filters or feature maps to illustrate how CNNs process time-frequency representations to identify speech patterns.

2.4.6 Advantages

CNNs leverage shared weights and local connectivity, reducing the number of parameters and computational cost compared to fully connected networks. They excel Convolutional Neural Networks (CNNs) offer several advantages in the realm of deep learning. They excel in feature learning by automatically extracting hierarchical representations from data. Beginning with simple features in initial layers and progressing to complex patterns in deeper layers, CNNs effectively capture spatial hierarchies and local patterns. This capability makes them particularly suitable for tasks like image recognition, where identifying objects involves understanding spatial relationships within the image. CNNs have advantage of parameter sharing within convolutional

layers, reducing the number of model parameters and enhancing computational efficiency. They also support transfer learning, enabling pre-trained models to be fine-tuned for specific tasks with smaller datasets. Real-time processing is feasible due to optimizations like GPU acceleration, making CNNs suitable for applications requiring rapid data analysis across various domains, including computer vision and natural language processing. Their robustness to variations in input data, such as changes in orientation or lighting conditions, further underscores their versatility and effectiveness in handling diverse datasets. CNNs consistently achieve state-of-the-art performance in benchmarks, showcasing their prowess in complex tasks and their widespread adoption in research and industry [11] [12].

2.4.7 Disadvantages

Despite their strengths, CNNs also present several challenges and limitations. Training deep CNN architectures demands substantial computational resources, including high-performance GPUs, which can be expensive and restrict their accessibility. Overfitting is a common issue, especially with limited datasets, where CNNs may learn noise or specific details that do not generalize well to unseen data. The interpretability of CNNs poses another challenge due to their complex architectures and hierarchical feature learning processes. Understanding why a CNN makes specific decisions can be difficult, impacting trust and transparency in automated systems. Moreover, CNNs rely heavily on large volumes of labeled data for effective training, which may be scarce or costly to obtain in certain domains. Selecting optimal hyperparameters, such as learning rates and kernel sizes, requires extensive experimentation and fine-tuning, adding complexity to model development. While CNNs excel in capturing spatial hierarchies, they may struggle with tasks requiring precise spatial localization, such as boundary detection in images. Deploying CNN models in resource-constrained environments may also be impractical due to their computational demands, limiting their applicability in certain contexts. These considerations highlight the trade-offs and challenges in leveraging CNNs for deep learning applications, emphasizing the need for careful design and implementation to maximize their benefits while mitigating their limitations.in capturing spatial and temporal patterns efficiently from large datasets [14] [15].

2.5 Implementation of the System

2.5.1 TensorFlow

TensorFlow is a robust open-source machine learning framework created by Google, aimed at streamlining the development, training, and deployment of machine learning models across multiple platforms. At its core, TensorFlow revolves around the concept of tensors, which are multi-dimensional arrays that represent data. These tensors flow through a computational graph,

where nodes represent mathematical operations, enabling efficient execution and optimization of complex computations.

In our project, TensorFlow serves as the backbone for developing a voice recognition system using deep learning techniques. TensorFlow's flexibility allows us to construct and train neural networks capable of recognizing spoken commands from audio samples. We utilize TensorFlow's high-level APIs like Keras for defining and configuring neural network architectures effortlessly. This includes stacking layers, specifying activation functions, and configuring optimization algorithms to improve model accuracy over time.

2.5.2 Dataset Description

The dataset used in this project consists of 25,000 and 1000 audio samples categorized into nine command classes: "forward," "backward," "go," "stop," "left," "right," "on," "off," and an additional command class. Our Urdu language data set consists of approximately 1000 voice samples, categorized into 5 commands:بائیں, آگے, پیچھے, دائیں , رکو each audio sample corresponds to one of these spoken commands, crucial for training and evaluating the voice recognition system.

2.5.3 Data Acquisition and Preprocessing

Audio samples are sourced from diverse environments and speakers to ensure variability and representative training data. Manual labeling of each audio file with its respective command class facilitates supervised learning. Ensuring a balanced distribution of samples across all nine classes to avoid model bias during training.

2.5.4 Data Preprocessing

Utilization of TensorFlow's tf.audio.decode_wav to convert audio files into numerical waveforms. Standardization of audio waveforms to a consistent amplitude range to maintain uniform input quality.

2.5.5 Model Architecture Overview

2.5.5.1 Input Layer

The input to the model consists of spectrogram images generated from audio waveforms. Spectrograms are 2D representations of audio signals where the x-axis frequencies and the y-axis represent frequency bins. Spectrograms are normalized using a Normalization layer in TensorFlow to ensure consistent scaling across different input samples.

2.5.5.3 Feature Extraction

A Resizing layer down samples the input spectrogram images to a consistent size suitable for processing by subsequent layers.

2.5.5.4 Convolutional Layers

Several convolutional layers (Conv2D) are employed to extract hierarchical features from the spectrogram images. These layers use small filters (kernels) to convolve over the input, capturing spatial patterns in the spectrogram.

2.5.5.5 Pooling and Regularization

After each convolutional layer, MaxPooling2D layers are used to down-sample the feature maps, reducing spatial dimensions and extracting dominant features. To prevent over fitting and improve generalization, Dropout layers randomly deactivate a fraction of neurons during training.

2.5.5.6 Flattening

The 2D feature maps extracted by convolution and pooling are flattened into a 1D vector to be fed into dense layers.

2.5.5.7 Dense Layers

Dense (Fully Connected) Layers: One or more dense layers (Dense) follow the flattened layer to learn higher-level features and make predictions. Each neuron in these layers is connected to every neuron in the previous layer.

2.5.5.8 Output Layer

The final dense layer has units equal to the 9 classes, representing the probabilities of each class (command). Typically, the output layer uses a softmax activation function to produce class probabilities that sum to 1.

2.5.6 Model Summary

Table. 2.1. Model Specs

Layer (type)	Output Shape	Param #
Resizing	(None, 32, 32, 1)	0
Normalization	(None, 32, 32, 1)	3
Conv2D	(None, 30, 30, 32)	320
Conv2D	(None, 28, 28, 64)	18496
MaxPooling2D	(None, 14, 14, 64)	0
Dropout	(None, 14, 14, 64)	0
Flatten	(None, 12544)	0

Dense	(None, 128)	1605760
Dropout	(None, 128)	0
Dense	(None, 9)	1161

2.5.7 Training and Evaluation

- Adoption of the Adam optimizer for gradient-based optimization during model training.
- Sparse categorical cross-entropy used to measure the difference between predicted and actual distributions.
- Evaluation metrics such as accuracy employed to gauge model performance during training and validation.

2.5.8 Training Process

- Training conducted over multiple epochs with early stopping to prevent over fitting.
- Optimization achieved through batch processing of 64 samples per iteration.
- Regular validation against a separate dataset to monitor model performance and prevent overfitting.

2.5.9 Transfer Learning for Urdu Voice Commands

- Collected and preprocessed an Urdu voice commands dataset, applying STFT and augmentation techniques.
- Retained pretrained convolutional layers for feature extraction and fine-tuned dense layers for Urdu command classification.
- Adapted the model to recognize Urdu-specific phonetic characteristics through training on the
- Trained the model with early stopping and evaluated performance using accuracy and confusion matrix on a test set.

2.6 Integration with Hardware

2.6.1 Components

- Raspberry pie 4
- Battery
- Motors
- Sensors
- Microphone

- Chassis

2.6.1.1 Raspberry pie 4

The Raspberry Pi 4 is a compact, powerful, and versatile single-board computer designed for a wide range of applications, from educational projects to complex industrial automation. It features a quad-core ARM Cortex-A72 CPU, up to 8GB of RAM, dual-band Wi-Fi, Bluetooth 5.0, and Gigabit Ethernet, significantly improving processing power and connectivity over its predecessors. The Raspberry Pi 4 supports dual 4K displays via its two micro-HDMI ports, making it suitable for multimedia and graphical applications. Its GPIO (General Purpose Input/Output) pins enable easy interfacing with sensors, motors, and other electronic components, making it a popular choice for DIY projects and prototyping. It has following characteristics.

- High Performance: The quad-core ARM Cortex-A72 CPU and increased RAM options (up to 8GB) offer substantial performance improvements for multitasking and running complex applications.
- Connectivity: Enhanced networking capabilities with Gigabit Ethernet, dual-band Wi-Fi, and Bluetooth 5.0 facilitate fast and reliable data transfer and wireless communication.
- Multimedia Support: Dual micro-HDMI ports supporting 4K resolution allow for high-quality multimedia output, ideal for media centers, digital signage, and gaming.
- Versatility: The GPIO pins and a wide range of compatible peripherals make the Raspberry Pi 4 suitable for numerous applications, including robotics, Iota, and home automation.
- Energy Efficiency: Low power consumption makes it an energy-efficient choice for continuous operation in embedded systems and portable projects.
- Cost-Effective: It offers high-end computing capabilities at an affordable price, making advanced technology accessible for educational purposes and budget-conscious developers.
- Community and Support: A large, active community provides extensive resources, tutorials, and forums, aiding both beginners and experienced users in their projects.

2.6.1.2 Battery

In the modern world, batteries are a necessity for everything from little electronic gadgets like watches and hearing aids to massive machines like cars and airplanes. This essay will examine batteries, including the various types, uses, and effects they have on our daily life.

There are various varieties of batteries, each having special qualities and uses. Some of the most popular battery kinds are listed here. Compared to most other battery types now on the market, Li-ion batteries have a high energy density. The most popular rechargeable batteries are lithium-ion batteries. They are found in many different types of equipment, including as power tools, electric cars, laptops, and cellphones. The advantages of lithium-ion batteries include their portability, high energy density, and several recharges.

Advantages

- Lithium batteries have a high energy density, which allows them to store a lot of energy in a relatively small space. They are therefore perfect for use in portable electronics like electric cars, computers, and cellphones.
- Long Life: Compared to most other battery types, lithium batteries have a longer lifespan. They can survive for several years before needing to be replaced, which over time makes them a cost-efficient option.
- High Voltage: Lithium batteries are appropriate for use in high-powered devices like power tools and electric cars because of their high voltage output.
- Relatively modest weight: Lithium batteries are perfect for use in portable devices where weight is an issue.

Disadvantages

- Expensive: Lithium batteries are more costly than other battery types, which may make them an unaffordable choice for some customers.
- Fire Risk: If damaged or overheated, lithium batteries may catch fire. Although this risk is reduced by the inclusion of safety features such protection circuits, it is still something to be concerned about.
- Temperature Sensitivity: The performance and longevity of lithium batteries may be impacted by temperature variations.
- Environmental Impact: If lithium batteries are not properly disposed of, they may have a detrimental effect on the environment. They contain elements that may be hazardous to the environment, such as lithium, cobalt, and nickel.

2.6.1.3 Motor

Each motor style on the market today has its own unique set of benefits and drawbacks. Although Direct Current (DC) motors have been used for decades in a variety of applications, their popularity has skyrocketed in the past few years [14]. Direct current (DC) motors are by far the most popular type of motor used in low-power devices such as toys, tools, and small appliances. They use a magnetic field generated by a stationary part (the stator) to spin a moving part (the rotor).

Advantages:

- Since DC motors can generate a lot of torque at low RPMs, they are useful in many situations that call for powerful initial acceleration.
- Third, DC motors may be easily controlled with simple circuits, allowing for precise speed control over a large operating range.
- DC motors are easier to construct and maintain than other types of motors because of their straightforward construction.
- Fifthly, DC motors are frequently less expensive than other motor types, making them a viable option for a wide range of uses.

Disadvantages

- DC motors have a relatively short lifespan, therefore they may require regular maintenance, such as brush and part replacement.
- When the brushes of a DC motor encounter the commutator, a spark is created. This spark can cause electromagnetic interference and, in the case of explosive situations, be dangerous.
- Third, DC motors require frequent servicing, which may be both time consuming and costly.
- Brushes and bearings will need to be cleaned and replaced as part of this routine upkeep.
- Fourth, DC motors aren't ideal for high-speed uses since their maximum speed range is lower than that of other motor types.

2.6.1.4 Sensors

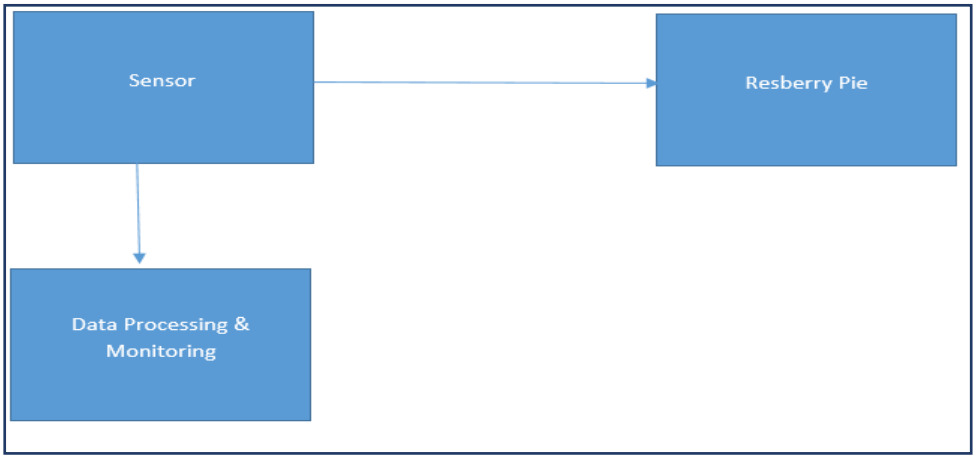


Fig. 2.6. Block Diagram of Temperature Sensor

2.6.1.4.1 Temperature Sensor

The temperature sensor is a critical component in our Voice Guardian system, designed to monitor the environmental temperature to ensure safe and optimal operation. By detecting temperature variations, the sensor helps prevent overheating of electronic components and provides crucial data for environmental monitoring in various applications, such as mining, healthcare, and industrial settings.

Temperature sensors measure the amount of heat energy in an environment and convert it into a readable form. The sensor typically works by utilizing materials whose electrical properties change with temperature. Common types of temperature sensors include thermocouples, resistance temperature detectors (RTDs), thermistors, and semiconductor-based sensors.

- Thermocouples **c**onsist of two different metals joined at one end, generating a voltage proportional to the temperature difference between the joined end and the other ends.
- RTDs made of pure metals whose resistance changes predictably with temperature.
- Thermistors semiconductor devices whose resistance decreases as temperature increases.
- Semiconductor Sensors use integrated circuits to measure temperature, providing digital or analogue outputs.

2.6.1.4.2 Toxic gas sensor

The MQ-135 Gas Sensor is a crucial component for detecting toxic gases in mining environments. Designed to detect a variety of harmful gases such as ammonia, nitrogen oxides, and benzene, this sensor operates by utilizing tin dioxide (SnO2) as its sensitive material. When toxic gases are present, the sensor's resistance changes, generating an electrical signal that can be interpreted by microcontrollers like the Raspberry Pi. This sensor's high sensitivity and rapid response times make it effective for early warning systems, providing timely alerts to mitigate potential hazards.

Its compact size and compatibility with the Raspberry Pi enable seamless integration into mobile robots, enhancing safety by continuously monitoring air quality in challenging mining conditions.

2.6.1.5 Suitability of Temperature Sensors for Mining Applications

Mining environments are often harsh and demanding, presenting significant safety risks. Monitoring temperature in these settings is crucial for preventing hazards, ensuring the safety of personnel, and maintaining the functionality of equipment. Temperature sensors are particularly suitable for mining applications because they provide real-time data on ambient and equipment temperatures.

The importance of temperature monitoring in mining cannot be overstated. Firstly, it plays a vital role in safety assurance by detecting abnormal heat levels that could indicate potential hazards such as equipment malfunctions or fire risks. Early detection of these issues allows for prompt intervention, reducing the likelihood of accidents and ensuring the safety of miners. Additionally, temperature sensors help protect equipment. Mining machinery operates under heavy loads and often in continuous cycles, which can lead to overheating and equipment failure. Temperature sensors provide early warnings of overheating, allowing for preventive maintenance and reducing costly repairs and downtime.

Moreover, environmental monitoring is another critical aspect. Temperature sensors help maintain safe and regulated working conditions, particularly in underground mining, where temperature control is vital for ventilation and worker safety. By ensuring that environmental conditions remain within safe parameters, temperature sensors contribute to a safer working environment. Operational efficiency is also enhanced through temperature monitoring. Maintaining optimal temperatures improves the efficiency of mining operations by better managing cooling systems and energy resources, leading to cost savings and enhanced performance.

Integrating temperature sensors into mining operations involves strategic placement in critical areas, such as near machinery, ventilation systems, and key structural points, to ensure comprehensive monitoring of the entire mining environment. Modern temperature sensors can transmit real-time data to a central monitoring station via wireless communication systems, enabling continuous surveillance and quick response to any irregularities. These sensors are designed to be compatible with existing mining infrastructure, including control systems and safety protocols, ensuring seamless implementation and operation without extensive modifications.

The robust design of these sensors allows them to withstand harsh conditions, including dust, moisture, and mechanical vibrations, ensuring reliable performance even in the most challenging environments. The use of temperature sensors in mining enhances the reliability of safety systems by providing accurate and consistent data, which is crucial for high reliability and operational safety. Preventive maintenance enabled by temperature monitoring reduces the risk of unexpected equipment failures, lowering maintenance costs and minimizing downtime. Continuous temperature monitoring ensures potential hazards are identified and addressed promptly, safeguarding the lives of miners. By optimizing cooling systems and maintaining ideal operating temperatures, mining operations can achieve higher efficiency and productivity.

In conclusion, temperature sensors are indispensable in mining applications due to their ability to enhance safety, protect equipment, and improve operational efficiency. By providing real-time temperature data and early warnings of potential hazards, these sensors play a crucial role in maintaining a safe and efficient mining environment. Integrating temperature sensors into the mining infrastructure is a strategic decision that ensures both the safety of personnel and the longevity of equipment, ultimately contributing to the overall success of mining operations.

2.6.1.6 Microphone

For embedding on a robot with a Raspberry Pi, the ReSpeaker USB Mic Array v2.0 is an ideal choice. This compact and high-quality microphone array is specifically designed for voice recognition applications. It features four high-performance microphones arranged in an array, which enhances its ability to capture voice commands from different directions and distances. The built-in beam forming technology focuses on the sound source, effectively isolating voice commands from ambient noise, ensuring accurate voice recognition even in noisy environments. Advanced noise suppression algorithms reduce background noise, while echo cancellation features eliminate echoes, ensuring clear and intelligible voice commands. The USB plug-and-play connectivity allows for easy integration with the Raspberry Pi without the need for additional drivers or software. The microphone array is designed for far-field voice recognition, making it ideal for mobile robots that need to respond to commands from several meters away. Its compact and lightweight design makes it easy to embed on a mobile robot without adding significant weight or bulk. Overall, the ReSpeaker USB Mic Array v2.0 enhances the functionality and reliability of Voice Guardians, ensuring high-quality voice capture and accurate recognition in various environments.

2.6.1.7 Chassis

The chassis of a mobile robot serves as its structural foundation, supporting various components such as sensors, actuators, and the control system. It is typically made from materials like aluminum, steel, or durable plastics to withstand rugged environments encountered in applications such as mining. The design of the chassis influences the robot's mobility, stability, and load-bearing capacity, crucial for navigating uneven terrain and carrying payloads. Integrated features like mounting brackets and compartments facilitate the installation and protection of electronic components, ensuring reliability and longevity in harsh conditions. Overall, a robust chassis is essential for the performance, durability, and safety of mobile robots deployed in challenging environments like mining.

2.6.2 Integration in the System

In our project, we use sensors for accurate and reliable readings. The sensor is interfaced with the Raspberry Pi 4, which processes the temperature data and uses it to make informed decisions about the robot's operation.

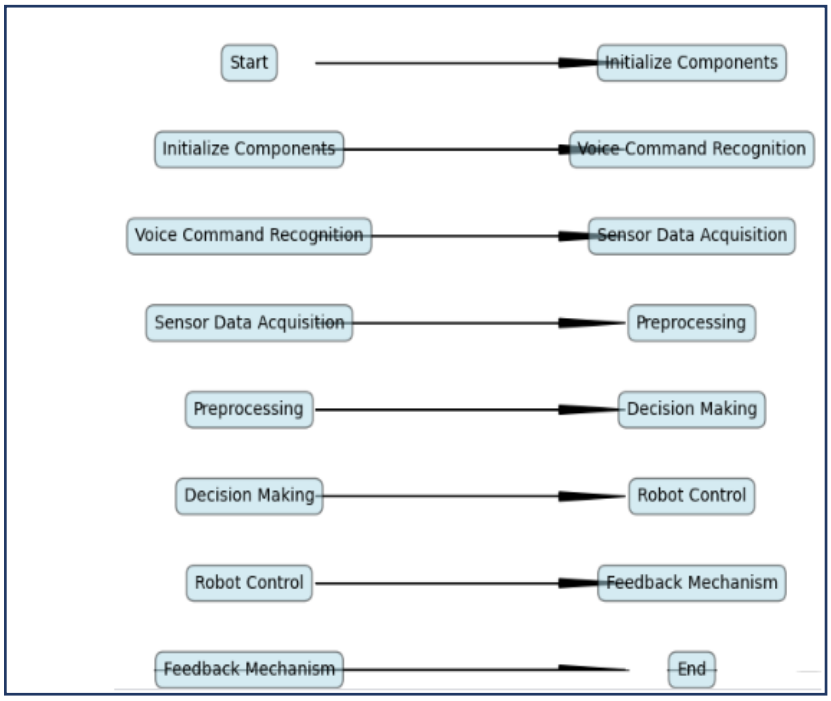


Figure. 2.9. Top Level System Integration Flow

2.6.3 Software Integration

Implementation of software components for seamless communication between the voice recognition system and the robot's onboard processor.

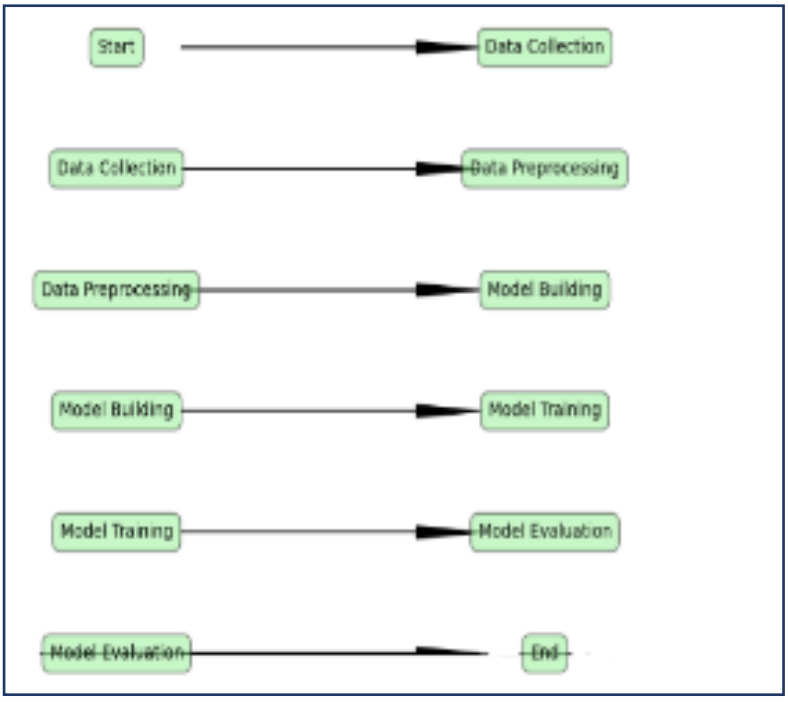


Fig. 2.6. Top-Level Model Integration Flowchart

CHAPTER 3

RESULTS AND DISCUSSIONS

3.1 Preprocessing and Spectrograms

In the pre-processing stage, raw audio signals were converted into spectrograms using Short-Time Fourier Transform (STFT). This transformation was essential for feeding the data into the Convolutional Neural Network (CNN), as spectrograms provide a visual representation of the signal's frequency content over time. Figures 6 illustrate the waveform and corresponding spectrogram for sample audio signals.

The pre-processing pipeline included data augmentation techniques such as adding noise, time-shifting, and pitch alteration to enhance the dataset's diversity. These techniques helped in improving the model's robustness by exposing it to a wider range of variations in the input signals.

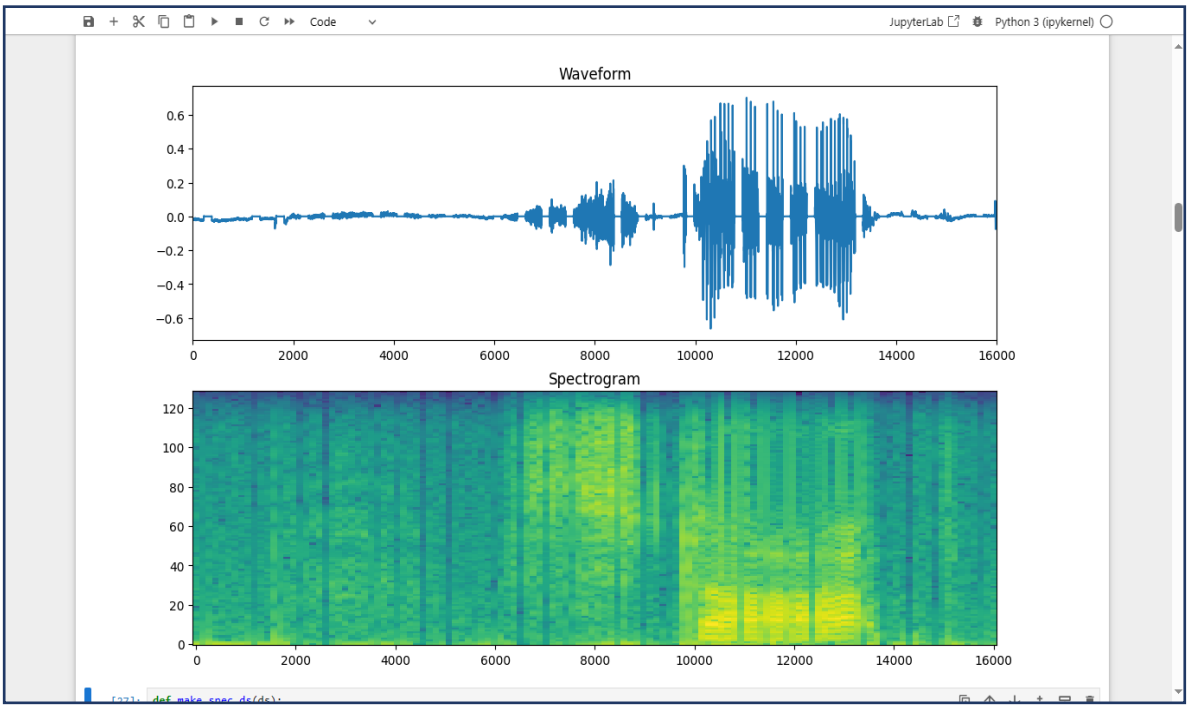


Fig. 3.1. Spectrogram Analysis of 'LEFT'

3.2 Model Performance

Following pre-processing, the CNN model was trained and evaluated on the processed spectrograms. The model achieved an impressive accuracy of 95%, demonstrating its strong capability in accurately identifying spoken words. Performance metrics such as precision, recall, and F1-score were calculated for each class of spoken words and are summarized in Figure 9. These metrics highlight the model's robust performance across various categories.

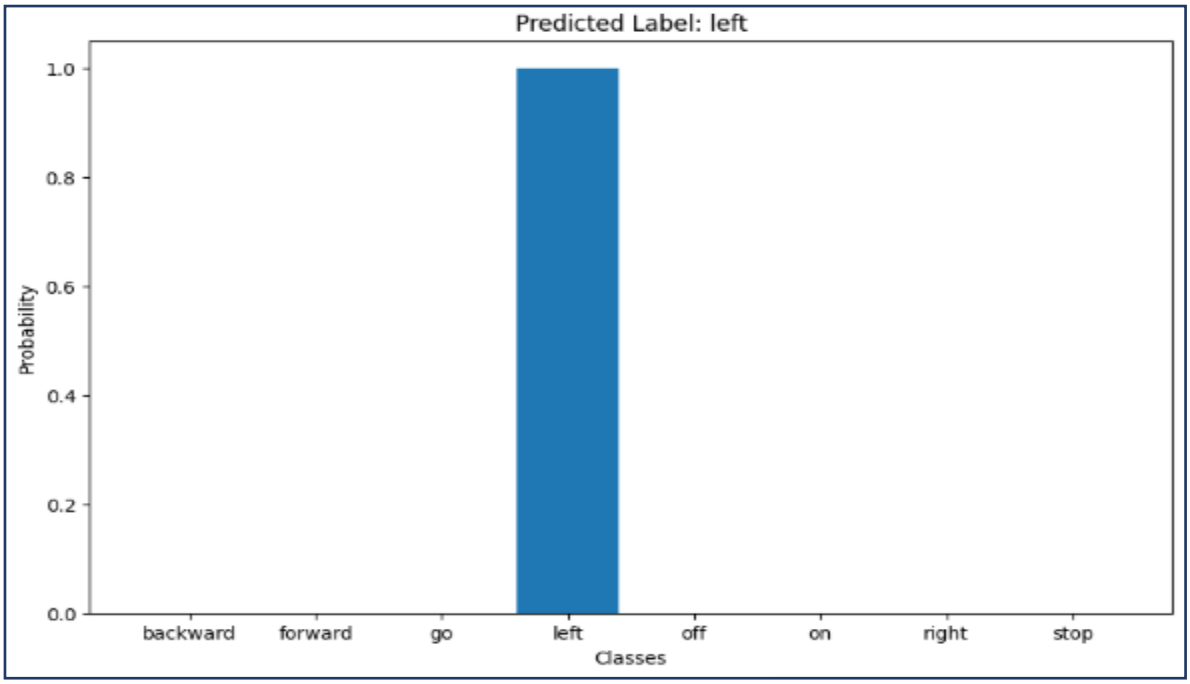


Fig. 3.2. Model Prediction

3.2.1 Training and Validation Loss

The training and validation loss curves over 15 epochs are shown in Figure 8. The training loss consistently decreased, indicating effective learning, while the validation loss plateaued after a certain number of epochs, suggesting that the model reached its optimal performance. The minimal gap between the training and validation loss curves suggests effective control over overfitting.

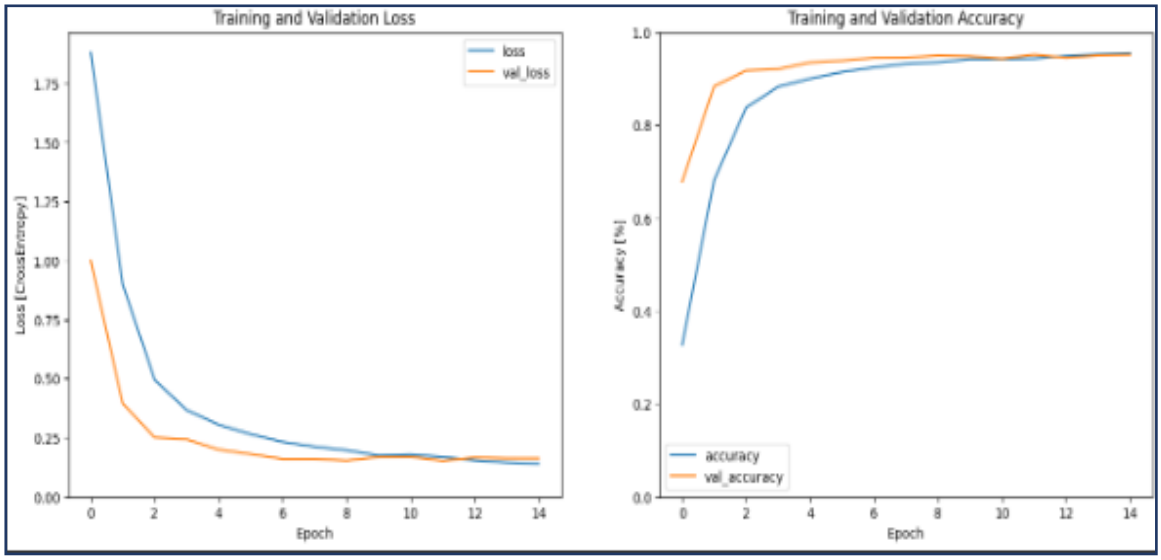


Fig. 3.3. Training and Validation

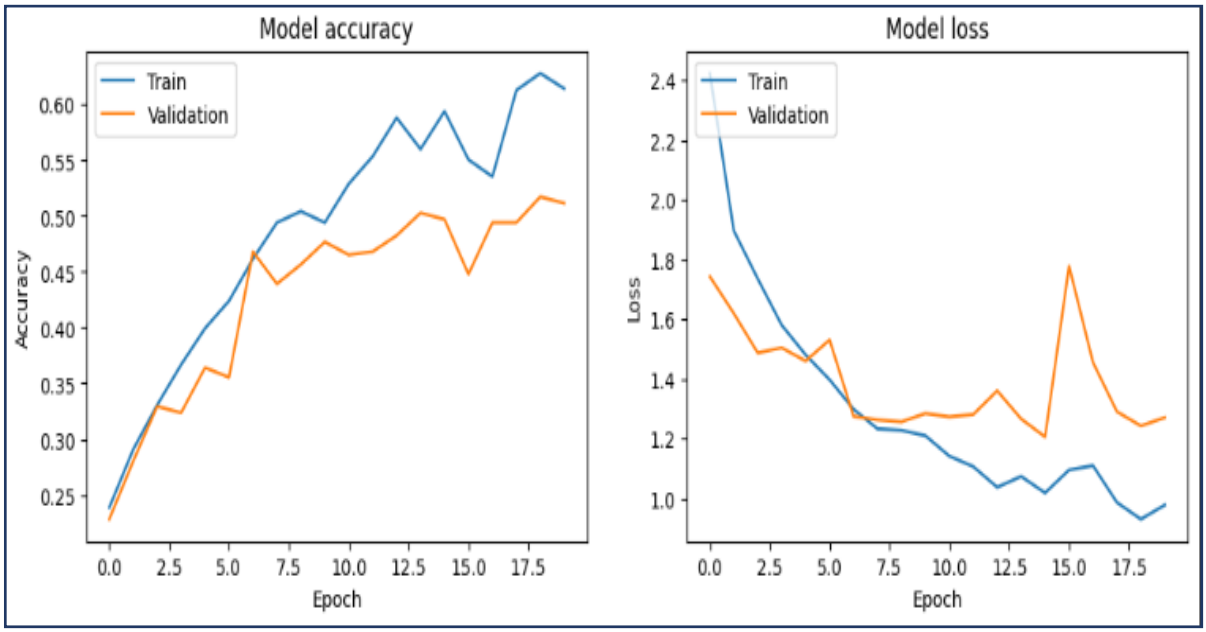


Fig. 3.4. Training and Validation (Urdu)

	precision	recall	f1-score	s
backward	0.94	0.92	0.93	
forward	0.96	0.88	0.92	
go	0.96	0.96	0.96	
left	0.96	0.95	0.96	
off	0.92	0.95	0.93	
on	0.93	0.95	0.94	
right	0.98	0.96	0.97	
stop	0.96	0.98	0.97	
accuracy			0.95	

Fig. 3.5. Score Table

```
11/11 [==============================] - 0s 2ms/step
Classification Report:
              precision    recall  f1-score   support

          91       0.65      0.55      0.78       پیچھے
          62       0.36      0.45      0.29       آگے
          68       0.39      0.49      0.33       بائیں
          66       0.33      0.23      0.62       دائیں
          59       0.57      0.59      0.56       روکو

    accuracy                           0.47       346
   macro avg       0.52      0.46      0.46       346
weighted avg       0.54      0.47      0.47       346
```

Fig. 3.6. Score Table (Urdu)

3.2.2 Confusion Matrix Analysis

The confusion matrix in Figure 10 provides a detailed view of the model's performance in individual classes. Most classes showed high true positive rates, although there were occasional misclassifications, such as confusing "backward" with "forward". This could be due to their similar spectral patterns. The Urdu language dataset shows average positive rates as dataset limitations Overall, the confusion matrix indicates strong performance, with most predictions aligning along the diagonal.

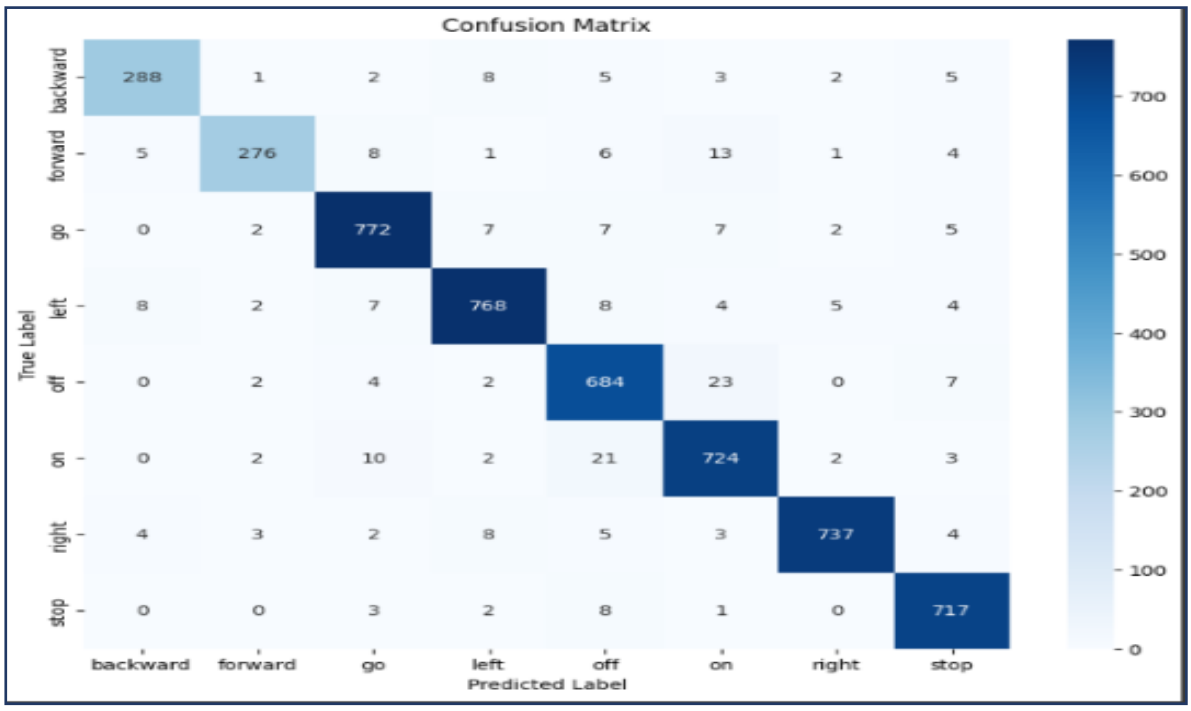


Fig. 3.7. Confusion matrix

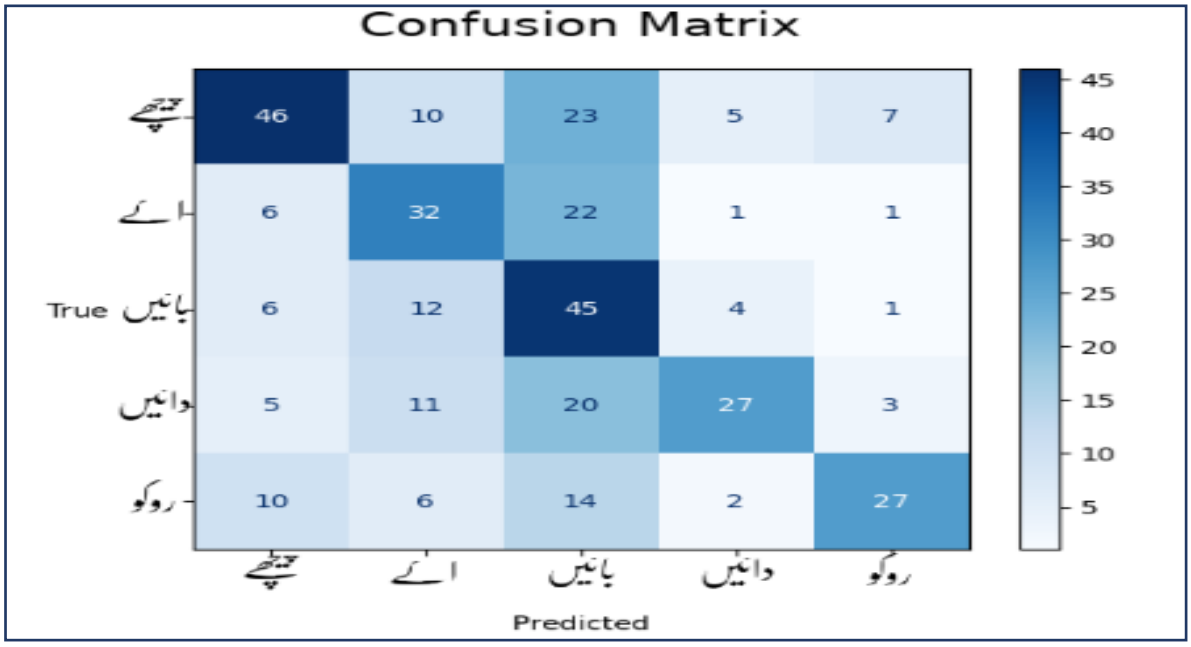


Fig. 3.8 Confusion matrix (Urdu)

3.3 Discussion

The results confirm that Convolutional Neural Networks (CNNs) are highly effective for voice recognition tasks. The ability of CNNs to learn hierarchical features from spectrograms allows them to capture intricate patterns in speech signals. This is a significant advantage over traditional methods like Mel-Frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC), which rely on manually crafted features. The CNN model's high accuracy and robust performance metrics demonstrate its superiority in handling the variability and complexity of spoken words.

Data augmentation techniques such as adding noise, time-shifting, and pitch alteration were applied to increase the diversity of the training dataset. This strategy improved the model's generalization capabilities, as evidenced by the minimal gap between training and validation losses. Augmentation helped the model to learn more robust features, making it less prone to overfitting and better at handling real-world variability in speech input.

While traditional methods like MFCC and LPC have been the cornerstone of speech recognition systems, they have limitations in capturing temporal dynamics and complex spectral patterns. CNNs, on the other hand, excel in these areas by leveraging convolutional operations that preserve spatial relationships in spectrograms. The superior performance metrics of the CNN model in this study highlight its effectiveness in extracting and utilizing these features for accurate voice recognition.

Training CNNs requires significant computational resources, particularly for deep architectures and large datasets. In this study, training was conducted on a high-performance GPU, significantly reducing training time. However, deploying such models in resource-constrained environments remains challenging. Techniques such as model pruning, quantization, and transfer learning can help mitigate these issues by reducing model size and computational requirements without substantially sacrificing accuracy.

Despite the high performance of the CNN model, some limitations were observed. Misclassifications between phonetically similar words indicate that further refinement of the model's architecture and training process is needed. Future work could explore more advanced network architectures, such as Residual Networks (ResNets) or Recurrent Neural Networks (RNNs), to improve temporal context understanding. Additionally, expanding the dataset to include more diverse speech samples and exploring unsupervised learning techniques could further enhance model robustness and accuracy.

CHAPTER 4
CONCLUSION, LIMITATIONS, AND RECOMMENDATIONS

4.1 CONCLUSION

The "Voice controlled mobile robot" project has successfully showcased the integration of advanced speech recognition algorithms and machine learning techniques to control a mobile robot using voice commands. This innovative system significantly enhances user experience by making robot control more intuitive and accessible, with a particular focus on improving safety in mining environments.

The project employs a high-quality microphone to capture voice commands from the user. These audio signals are then processed in real-time using the Short-Time Fourier Transform (STFT) technique, which converts the raw audio into spectrograms. These spectrograms, which provide a visual representation of the signal's frequency content over time, are crucial for the subsequent machine-learning processes.

The transformed data is fed into Convolutional Neural Networks (CNNs), which are designed to accurately predict the voice commands. CNNs are particularly well-suited for this task due to their ability to recognize patterns in spectrogram images, allowing them to differentiate between various voice commands based on their spectral features.

One of the key achievements of this project is its implementation on a Raspberry Pi 4. This demonstrates that advanced machine learning algorithms can be deployed on small, affordable hardware while still maintaining high levels of efficiency and accuracy. The use of Raspberry Pi 4 underscores the potential for widespread adoption of this technology in various practical applications, especially where cost and portability are critical considerations.

By combining STFT, spectrogram analysis, and CNNs, the project not only achieves real-time voice command recognition but also opens up new possibilities for enhancing safety and operational efficiency in hazardous environments like mines. The ability to control robots using voice commands can significantly reduce the need for manual operation in dangerous conditions, thereby improving safety for workers.

Overall, the "Voice controlled mobile robot" project demonstrates the feasibility and effectiveness of integrating machine learning techniques with voice recognition for practical, real-world applications. The success of this project highlights the potential for similar systems to be developed

and deployed across various industries, offering intuitive and accessible control solutions that enhance both safety and productivity.

4.2 LIMITATIONS

The project encountered several limitations that influenced its performance and potential scalability. Firstly, the Convolutional Neural Network (CNN) model used, while generally accurate, occasionally struggled with misclassifying commands, especially those with similar spectral patterns. This issue underscores the necessity for more sophisticated models or an increase in training data to enhance the model's ability to distinguish between closely related commands effectively.

Secondly, the project's reliance on the Raspberry Pi 4 introduced hardware constraints that could impact future applications. Although the Raspberry Pi 4's processing capabilities were adequate for this project's scope, they may restrict the system's scalability and the complexity of operations in more advanced applications. Adopting more powerful hardware could facilitate the implementation of more intricate functionalities and support larger datasets.

Moreover, environmental noise posed a significant challenge to the voice recognition system's performance. High levels of ambient noise in real-world environments, such as mining sites, could interfere with accurate command detection. Implementing advanced noise-cancellation techniques or utilizing robust microphones designed for noisy environments could potentially mitigate these issues and improve the system's reliability under adverse conditions.

Additionally, the project faced limitations related to the availability of data. The dataset used for training the model was relatively small, which can constrain the model's ability to generalize effectively across a broader range of conditions and commands. Increasing the volume and diversity of training data could address this limitation and enhance the system's robustness and accuracy over time.

Lastly, the system's input method, based on short audio snippets for command recognition, also presented challenges. Short utterances can sometimes lack context or clarity, making accurate interpretation more challenging. Addressing this limitation might involve exploring techniques to capture longer, more informative input segments or incorporating context-aware processing algorithms to improve command understanding and system responsiveness.

4.3 FUTURE WORKS

The future work for the Voice Guardian project focuses on several key areas of improvement to enhance its functionality, adaptability, and application scope. First, advancements in model

sophistication are crucial. Implementing state-of-the-art machine learning algorithms, such as hybrid models combining Convolutional Neural Networks (CNNs) with Recurrent Neural Networks (RNNs), will significantly improve the accuracy and robustness of voice command recognition. Additionally, expanding the dataset with diverse voice samples, including various accents and speaking speeds, will enable the model to better generalize across different user inputs.

Hardware enhancements play a pivotal role in extending the capabilities of the robot. Upgrading to more powerful processors or specialized hardware like GPUs and TPUs will bolster real-time processing capabilities, facilitating the implementation of more complex algorithms and reducing response times for voice commands. Integrating advanced sensors, such as high-fidelity microphones and environmental monitoring devices like LIDAR and toxic gas sensors, will improve the robot's functionality in diverse environments, enhancing environmental perception and navigation accuracy.

Optimizing battery life through better power management and higher-capacity batteries will extend operational duration, making the robot more suitable for prolonged field use. Application expansion is another critical area. Enhancing safety features, such as detecting hazardous gases and structural instabilities in mining environments, will strengthen safety protocols and protect personnel. In industrial settings, deploying the voice-controlled system in dangerous or remote areas will streamline operations and boost productivity.

Moreover, integrating the system with smart home devices for home automation and personal assistance will offer hands-free control for everyday tasks, benefiting users, including those with disabilities. This comprehensive approach aims to advance the Voice Guardian project, making it more versatile, efficient, and applicable across a wide range of real-world scenarios and industries.

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